MIL-S-28856A(EC)
6 August 1982
SUPERSEDING
MIL-S-28856(EC)
19 December 1980
(See 6.4)

MILITARY SPECIFICATION

SYNCHRONIZER, VIDEO, SN-501/UPX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, performance, test, manufacture, and acceptance requirements for the SN-501/UPX Video Synchronizer, hereinafter referred to as the equipment, for Air Traffic Control Radar Beacon System (ATCRBS) Identification, Friend or Foe (IFF) Mark XII System (AIMS) interrogators.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-T-28735	Test Set, Radar, AM/UPM-137
MIL-G-28744	Generator, AIMS Pretrigger, SG-841( )/UPX
MIL-C-39012	Connector, Coaxial, Radio Frequency; General Specification For

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications, Standards, And In Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definition Of And Basic Requirement For Enclosures For Electric And Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-461	Electronic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurment Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-721	Definitions Of Effectiveness Terms For Reliability, Maintainability, Human Factors, And Safety
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 070-1	Interface Standard For Shipboard Systems, Section 070-Part 1 D.C. Magnetic Field Environment (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MS-3102	Connector, Receptacle, Electric, Box Mounting

HANDBOOK

MILITARY

MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

(Copies of specifications, standards, and handbooks required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70-1978	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General. The equipment is designated as shipboard equipment and shall be in accordance with MIL-E-16400, except as hereinafter specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design requirements.

3.2.1 General design. The equipment shall provide the time coordinating capability to permit operation of the interrogator either independently of radar synchronization (free-running), or slaved to the radar synchronization. There are two basic system interfaces that shall be satisfied by the design of the equipment. The first system (see FIGURE 1) is the operation of one interrogator with one radar system with the interrogator either slaved to the radar pulse repetition frequency (PRF) or operating independently at some PRF other than that of the radar. The second system (see FIGURES 2 and 3) is the operation of one interrogator simultaneously with two radar systems having different PRFs. To accomplish the second system, it will be acceptable if the interrogator is slaved to one radar and thereby not slaved to the other. However, the equipment shall accept the PRF of the controlling (slaving) radar whether it is greater than, or less than, the other radar.

3.2.1.1 IFF video and triggers. The IFF video signals and the mode tags shall be accepted without modification from the interrogators. The performance of the equipment in buffering these signals as outputs shall not alter the signal characteristics of the inputs.

3.2.1.2 Radar triggers. The equipment shall interface with radar triggers of positive polarity (see 3.2.16), having constant, staggered, or randomly jittered repetition rates. These triggers will precede radar zero range by 60 microseconds (μs) to 440 μs to allow IFF video decoding time.

3.2.1.3 Functions. The equipment shall provide AIMS composite video reply signals and mode tags from the interrogator at display times of the associated video display of synchronized and nonsynchronized radar systems (up to two radar systems).

3.2.1.4 Interrogators. The equipment shall interface with the AN/UPX-23 and AN/UPX-27 interrogators.

3.2.2 Capability. As a minimum, the equipment shall provide the following capabilities:

a. Synchronize IFF to a radar that is operating independently of the IFF timing (see FIGURE 1).

b. Synchronize IFF video to two radars when one radar is operating independently and the other radar is triggering (slaving) the interrogator. The pulse repetition interval (PRI) of the slaving radar shall be greater than the intervals of the independently operating radar (see FIGURE 2).

c. Synchronize IFF video to two radars when one radar is operating independently and the other radar is triggering (slaving) the interrogator. The PRI of the slaving radar shall be less than, or varying from less than, to greater than, the interval of the independently operating radar (see FIGURE 3).

3.2.3 Modular construction. The equipment shall employ modular design to provide easily replaceable assemblies, and subassemblies. Each replaceable assembly or subassembly, including the power supply, shall provide complete circuit functions with readily accessible internal test points. Plug-in assemblies shall be of the printed circuitry type which can be removed without need of unsoldering and which can be easily repaired. Each printed circuit board (PCB) or plug-in assembly shall be individually keyed to prevent insertion into the wrong connector, or reversing. A warning notice shall be provided within the front panel stating disconnect power prior to removal or insertion of modules.

3.2.3.1 PCBs. All PCBs shall be designed and manufactured in accordance with MIL-STD-275, and subassemblies, repair parts, and other piece-part items.

3.2.3.2 Interchangeability. For like equipments, all parts, including removable assemblies, and subassemblies, repair parts, and other piece-part items, shall be interchangeable physically and electrically with corresponding items or parts without alteration or use of undue force. The equipment design shall include tolerances sufficient to accommodate the replacement of various articles such as transistors, resistors, capacitors, integrated circuits (ICs), and other parts involved having limited dimensions and without departure from specified performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

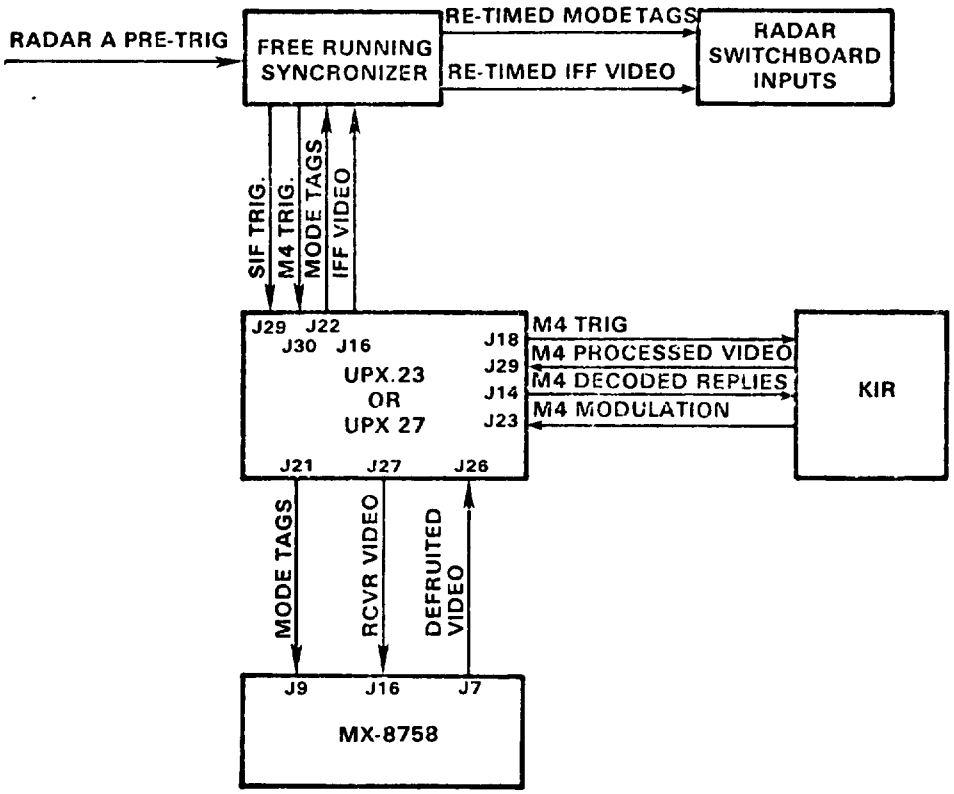


FIGURE 1. One radar system interconnection diagram for synchronizer when interrogator is independent of or slaved to PRF of radar.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

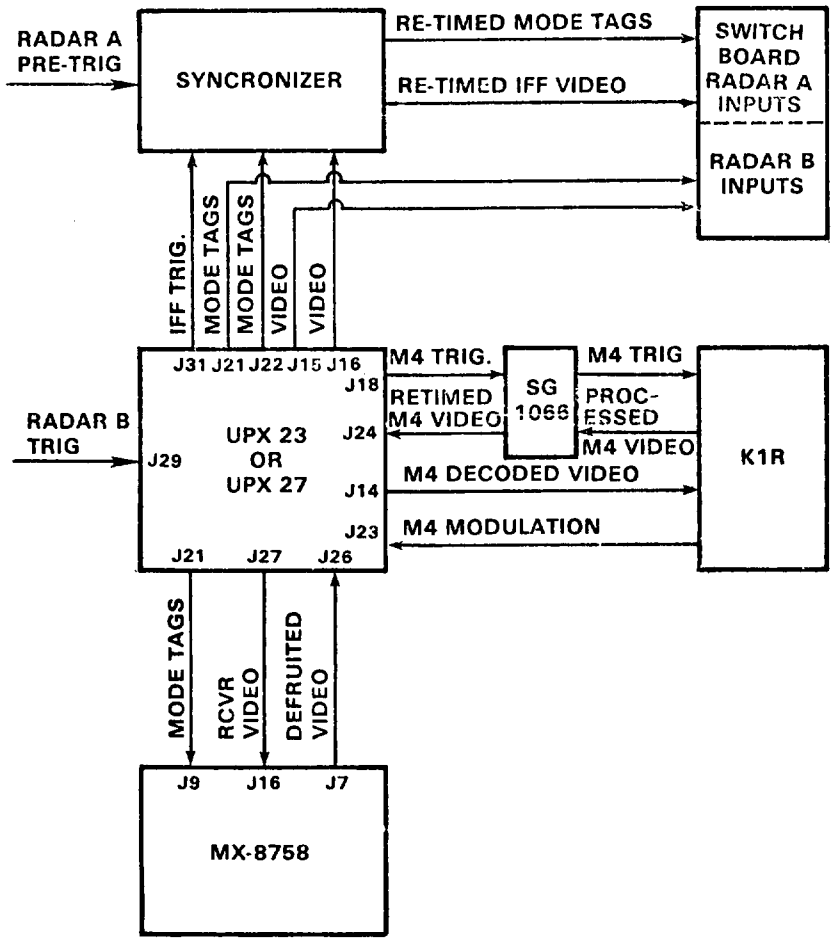


FIGURE 2. Two radar system interconnection diagram for the equipment when interrogator is slaved to lower PRF radar (Radar B).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

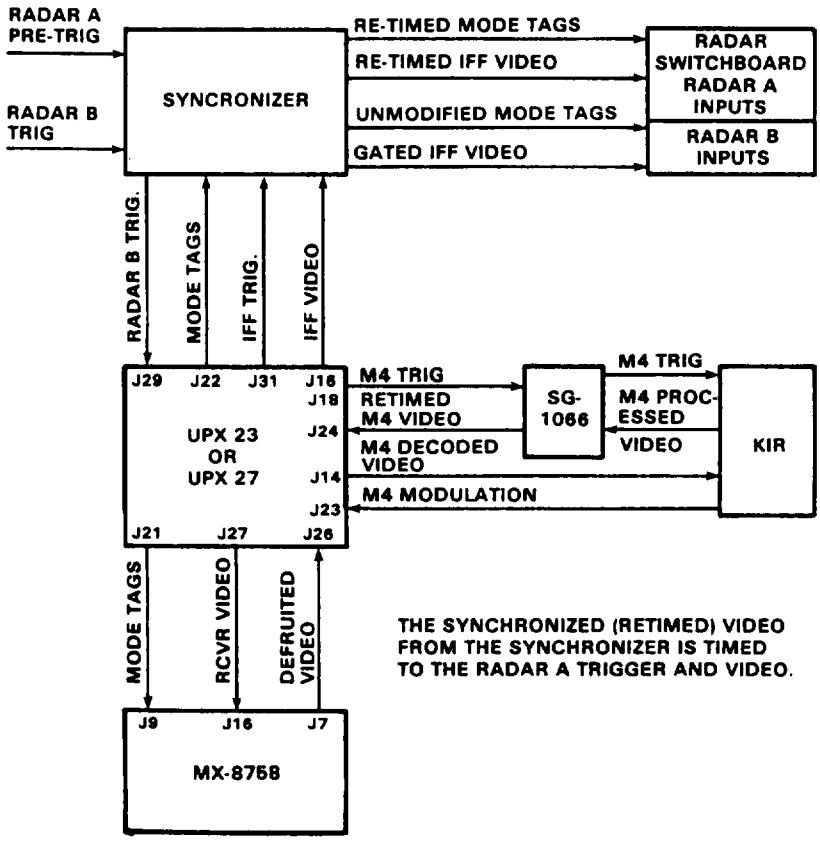


FIGURE 3. Two radar system interconnection diagram for the equipment when interrogator is slaved to lower PRF radar (Radar B).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.3.3 Encapsulation and embedment. Except for the power supply module, encapsulation or embedment of modules, assemblies and subassemblies shall not be permitted. If an encapsulated or embedded power supply is selected for use in the equipment, then the specific power supply shall be approved by the procuring activity.

3.2.4 Composition. The composition of the equipment shall be as specified in 3.2.4.1 through 3.2.4.3.

3.2.4.1 Active devices. All active circuits (amplifiers, memory storage, power supply, and so forth) used in the equipment shall employ solid-state devices (diodes, transistors, ICs and so forth). Use of thermionic electron tubes shall not be permitted.

3.2.4.2 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.2.4.3 Fungus nutrient parts or materials. Fungus nutrient parts or materials shall not be used in the equipment.

3.2.5 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified herein, under the environmental conditions specified in 3.2.5.1 through 3.2.5.5 and MIL-E-16400.

3.2.5.1 Temperature. The operating and nonoperating environment temperature shall be Range 4, except the operating temperature limits shall be -28° Celsius (C) to 65°C.

3.2.5.2 Salt-fog (spray). The complete equipment shall withstand the effects of the salt-fog (spray) test of MIL-E-16400 for sheltered equipment.

3.2.5.3 Vibration. The vibration shall be as specified in MIL-E-16400, except the upper vibration frequency limit shall be 55 hertz (Hz).

3.2.5.4 Inclination. The equipment shall be capable of withstanding the inclination test limits of MIL-E-16400 for surface ship installations.

3.2.5.5 Magnetic field environment. The equipment shall be compatible with the magnetic field environmental interface constraints of Section 070-1 of DoD-STD-1399.

3.2.6 Mechanical design. The mechanical design of the equipment shall be as specified in 3.2.6.1 through 3.2.6.9.

3.2.6.1 General mechanical design. The equipment shall be housed as a single unit consisting of a chassis assembly, a front panel, a rear panel, and an equipment case. The chassis assembly shall be mechanically mounted to the equipment case.

3.2.6.1.1 Chassis assembly. The chassis assembly shall be composed of plug-in modules and a chassis frame. Access of the equipment interior shall be accomplished by first releasing no more than six captive fastener devices located on the front panel surface and swinging the hinged front panel to the service position.

3.2.6.1.2 Equipment case. The case shall form the top, bottom, and two lateral sides of the equipment.

3.2.6.1.3 Rear panel. The rear panel of the equipment shall be a removable part of the equipment case and not part of the equipment chassis. Electrical connectors (with mating connectors on the chassis) shall be provided on the internal power and signal cables (conductors) coming from the rear panel to facilitate complete removal of the rear panel from the case. All signal input and output connectors shall be installed on the removable rear panel except for the test signal outputs which shall be mounted on the front panel.

3.2.6.2 Mounting provisions. The equipment case shall have mounting provisions on the bottom surface of the equipment case for direct mounting to a horizontal surface. The spacing of the mounting provisions and the mechanical configuration shall be identical to that of the SG-841 pulse generator in accordance with MIL-G-28744.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.6.3 Size. The maximum dimensions (including all provisions from the surfaces) of the equipment shall be as follows:

a. Height: 12.7 centimeters (cm) (5 inches (in))

b. Width: 26.29 cm (10.35 in)

c. Depth: 41.91 cm (16.5 in)

3.2.6.4 Weight. The weight of the equipment shall not exceed 11.79 kilograms (kg) (26 pounds (lbs)).

3.2.6.5 Cooling. The use of blowers or forced air as a cooling method shall not be permitted.

3.2.6.6 Degree of enclosure. The degree of enclosure shall be dripproof (45 degrees) in accordance with MIL-STD-108 as intended for installations within internal ship spaces.

3.2.6.7 Service and access. Accessibility shall be in accordance with MIL-E-16400, except that the equipment design shall permit easy and ready access to internal parts, assemblies, and subassemblies with the equipment in the service position.

3.2.6.7.1 Extender boards and cables. To facilitate ease of maintenance of the equipment, PCB extender cards shall be provided for displacing the PCBs and the power supply from their normal position.

3.2.6.8 Location and access to internal facilities and adjustments. Internal test points and jacks, functional switches and adjustments, and link settings shall be internally located, grouped in accordance with their functions to the greatest extent possible and shall be easily and readily accessible with the equipment in the service position.

3.2.6.8.1 Internal adjustments. Unless otherwise specified herein, all internal adjustments shall be continuously adjustable through their specified ranges.

3.2.6.9 Mating connectors. A mating power connector shall be provided in accordance with MIL-E-16400 to mate with the equipment rear panel power connector (see 3.2.13.1). No other mating connectors are required.

3.2.7 Maintainability.

3.2.7.1 Corrective maintenance (corresponds to organizational level). The equipment shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 20 minutes and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile), not exceeding 40 minutes when corrective maintenance is accomplished by replacement of lowest subassemblies (modules, printed wiring assemblies, and so forth) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The M_{ct} includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout. Built-in-test (BIT) circuits are included in the M_{ct} requirements. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.2.8 Quantitative reliability requirements. The specified upper test mean-time-between failures (MTBF) (θ_0 as defined by MIL-STD-781), shall be 4000 hours and the lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 2000 hours.

3.2.9 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein.

3.2.9.1 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. Abbreviations used on panels shall be in accordance with MIL-STD-12.

3.2.10 Self-test capability test points and general purpose electronic test equipment (GPETE). The requirements for self-test capability, internal test point criteria, and GPETE shall be as specified in 3.2.10.1 through 3.2.10.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.10.1 Self-test capability. A simple BIT facility shall be provided for checking the complete storage function (see 3.2.17.4) and via a built-in electronic counter, or similar device, shall be capable of indicating any defective storage cell. Operation of this test function shall be through a manually-operated momentary-off-momentary self-test switch mounted on the front panel and both the high and low state of the storage function shall be tested. Self-test shall only be entered upon momentarily depressing the self-test switch from the center-off position to test position. Upon finding a failure, the test shall stop and a light emitting diode (LED) display shall identify the memory chip and address at which the failure occurred. To advance the test, the self-test switch shall again be depressed to the test position and the test shall continue until the next fault is identified or a test complete indication is observed. Likewise in the case of multiple faults, advancement to each succeeding fault, shall be accomplished in this manner. By momentarily depressing the self-test switch to the escape position the test sequence shall be discontinued at any selected point during the self-test sequence, and the equipment automatically shall resume normal operation. A test complete indication (LED) shall be used to display the completion of the self-test and shall remain illuminated for approximately 10 seconds (s) to allow observation. Application of primary power or interruption of primary power shall not activate the self-test feature or otherwise prevent normal operation following resumption of primary power. A table in the instruction manual shall be sufficient for identifying the piece-part containing the defective storage cell.

3.2.10.1.1 Protection from BIT circuit failures. The ability of the equipment to perform its intended function shall not be impaired by the malfunction or failure of any part of the BIT circuits.

3.2.10.2 Internal test point criteria. Test points, test jacks, or both, shall be provided to permit the injection and measurement of signals, the monitoring of signals at the input/output (I/O) terminals, and for observation of prime signal paths for the assemblies and subassemblies. By means of these test points, test jacks, or both, fault isolation down to a singular module is attainable by use of the AN/UPM-137A in accordance with MIL-T-28735 and procedures to be specified in the technical manual.

3.2.10.2.1 Fault isolation. Test points, test jacks, or both, shall be provided to permit the injection and measurement of signals at I/O terminals, respectively, of assemblies and subassemblies for the purpose of isolating all fault to the individual electrical functions located within the assembly or subassembly. Test points and test jacks shall provide means for performing fault isolation on the assemblies and subassemblies when removed from the prime equipment.

3.2.10.2.2 Adjustments. Test points, test jacks, or both, shall be provided to permit adjustment of the equipment and the assemblies and subassemblies of the equipment.

3.2.10.3 GPETE. GPETE to be used at all levels of maintenance shall be selected from the Standard and Substitute standard equipments listed in MIL-STD-1364.

3.2.11 Workmanship.

3.2.11.1 Workmanship screen (random vibration). The equipment shall withstand a defect detection vibration screen of random type vibration of $0.04g^2/Hz$ ± 3 decibels (dB) from 80 Hz to 350 Hz (see 4.5.2.1).

3.2.11.2 Workmanship screen (temperature cycling). All equipment shall withstand a temperature cycling screen to ensure elimination of initial failures due to random defective components and marginal operation due to component lot variations or poor assembly processes and procedures (see 4.5.2.2).

3.2.12 Electromagnetic compatibility (EMC). The equipment shall be self-compatible, shall operate compatibly with the radars and interrogators for which it is intended, and shall conform to the following requirements of MIL-STD-461:

a. CE01 narrowband (NB), as shown in FIGURE 4

b. CE01 broadband (BB), not required

c. CE03 NB, as shown in FIGURE 5

d. CE03 BB, as shown in FIGURE 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

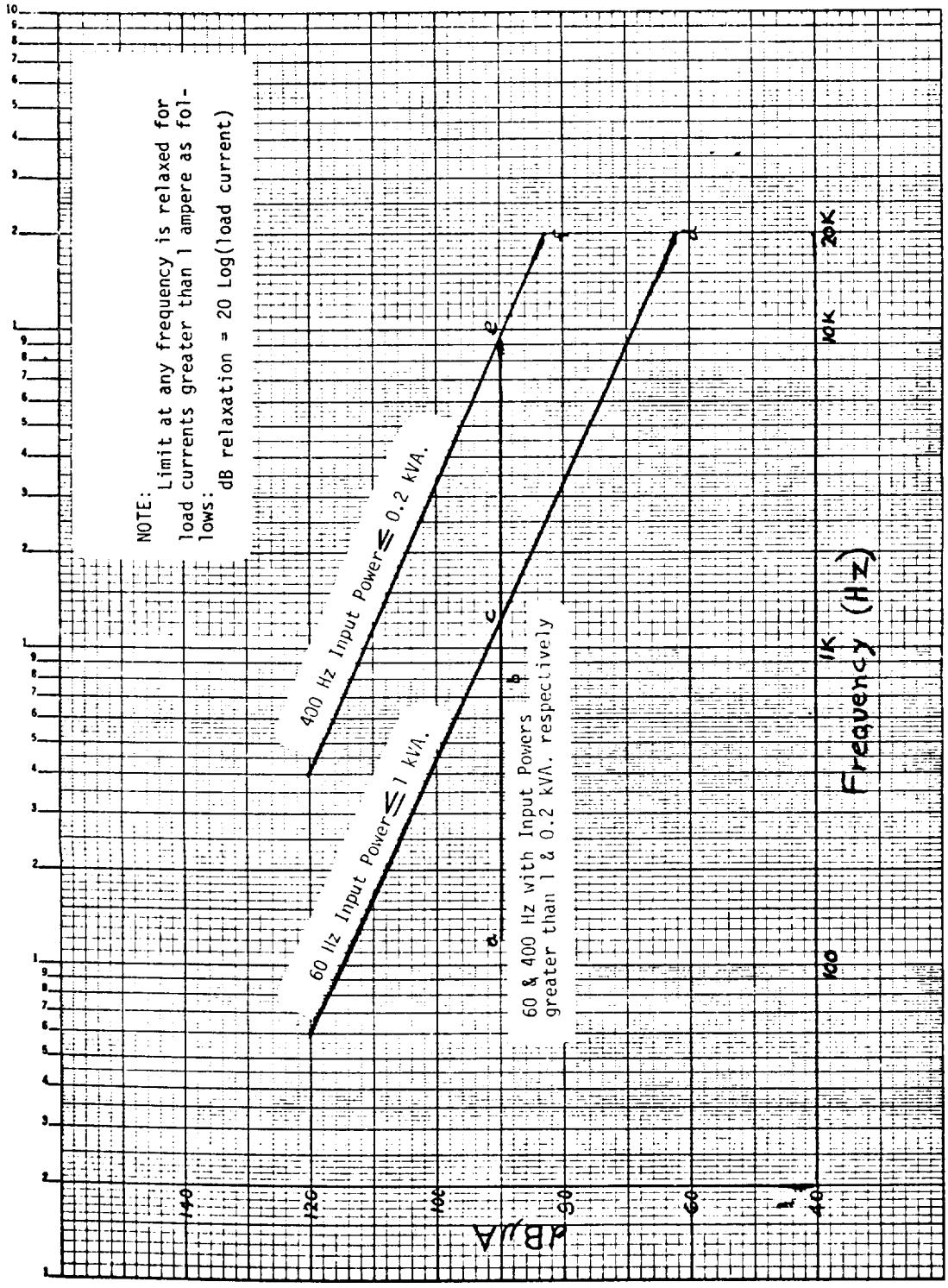


FIGURE 4. Limit for CE01 narrowband emissions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

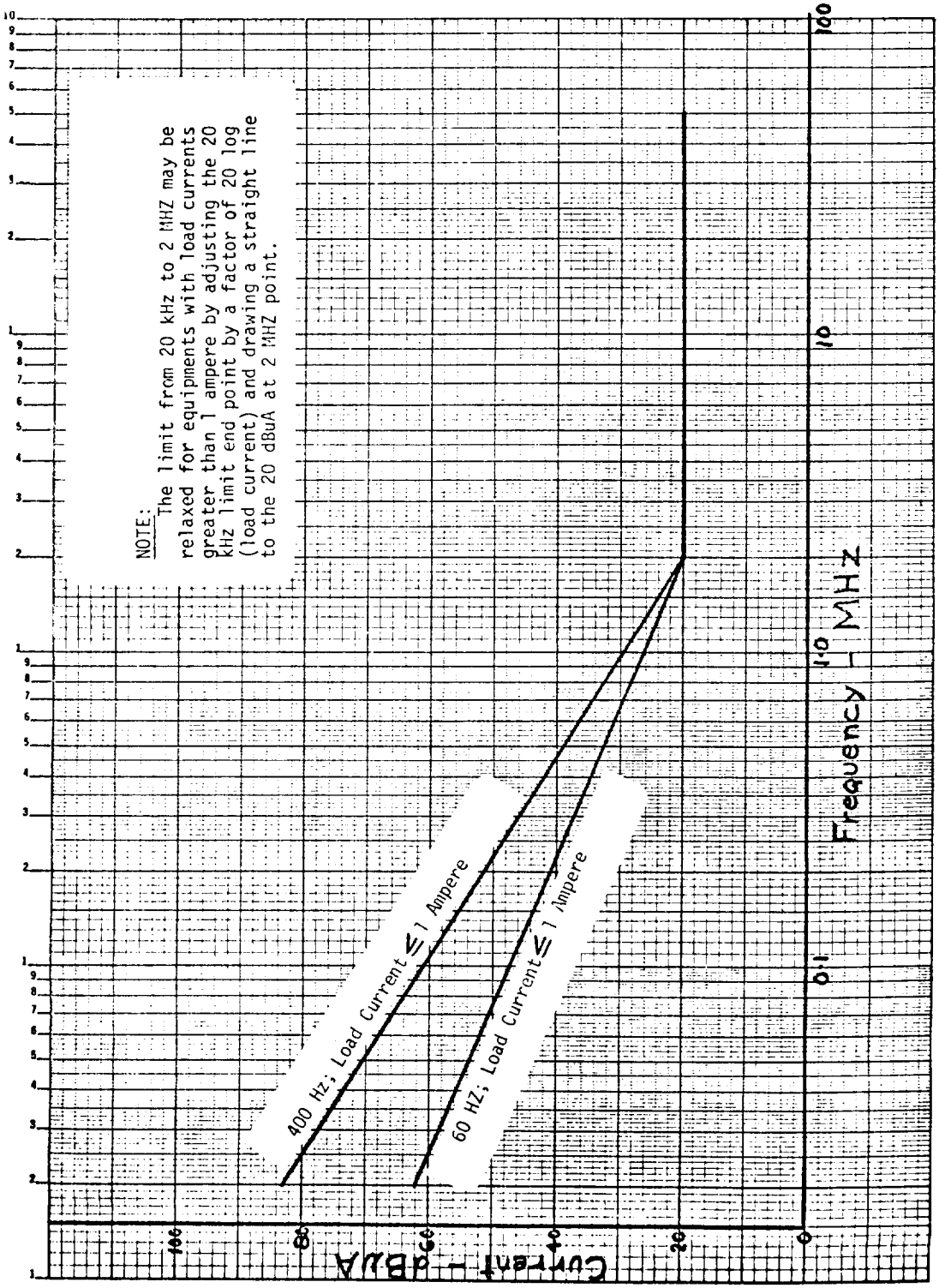


FIGURE 5. Limit for CE03 narrowband emissions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

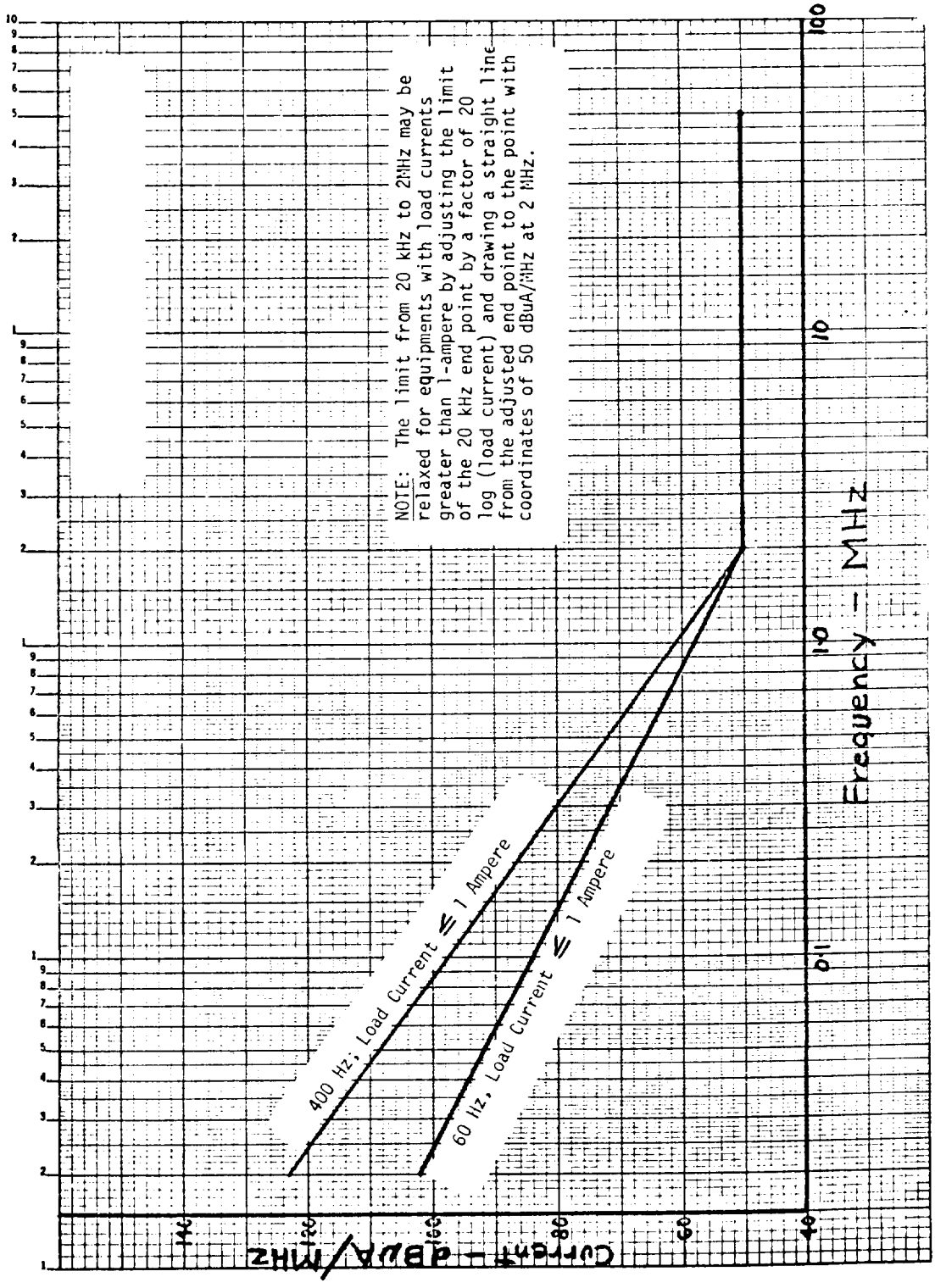


FIGURE 6. Limit for CE03 broadband emissions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

e. CS01, as shown in FIGURE 7

f. CS06, perform using a 400-volt (V) peak spike at 5 μs wide

g. RE01, as shown in FIGURE 8

h. RS01, as shown in FIGURE 8

i. RS03, perform at 10 volts per meter (V/m) from 14 kilohertz (kHz) to 10 gigahertz (GHz)

j. CE02, CE04, CS02, and RE02 shall be performed without change to limits.

3.2.12.1 Power supply electromagnetic interference (EMI) design. MIL-HDBK-241 may be used for guidance in EMI reduction in power supply design.

3.2.13 Shipboard alternating current (AC) power source. The shipboard AC power source shall be in accordance with MIL-E-16400, except the equipment shall be designed and constructed to operate from a 115-volt alternating current (VAC) ±10 percent, 57 Hz to 63 Hz, and 360 Hz to 440 Hz single-phase source. The 60 Hz and 400 Hz shall never be used simultaneously.

3.2.13.1 Power connector. The power connector shall be mounted on the rear panel and shall be in accordance with MS3102R-14S-7P, or equivalent. Equipment power connectors for 115 VAC input power and interconnecting AC power cable connectors between equipments shall have only three pins designated A, B, and C. Pin assignment and conductor color code within the equipment shall be as shown in TABLE I.

TABLE I. Pin assignment and conductor color codes.

Pin designation	Conductor assignment	Color code
A	115 VAC return	White or light blue
B	Safety ground	Green or green with yellow stripe
C	115 VAC high (hot)	Black or brown

Black and white color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.2.13.2 Power supply. The power supply used in the equipment shall not be damaged by either open or short circuiting any combination of the power supply outputs. Furthermore, any combination of the outputs of the power supply shall have at least a 20 percent current reserve capacity beyond that required for operation under any combination of conditions included in this specification.

3.2.14 Front panel. Test points (see 3.2.15.1), the equipment electrical power on-off switch, the power indicator lamp, and the overload protective device shall be mounted on the surface of the front panel.

3.2.14.1 Equipment electrical power on-off switch. A switch for disconnecting equipment from all electrical power systems shall be mounted on the equipment front panel and its function clearly labeled. It shall break all power conductors of the power circuit.

3.2.14.2 Power indicator lamp. A green or clear power indicator lamp shall be mounted near the switch to indicate when the equipment is energized. The lamp shall be connected to the load side of the switch and across the input power conductors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

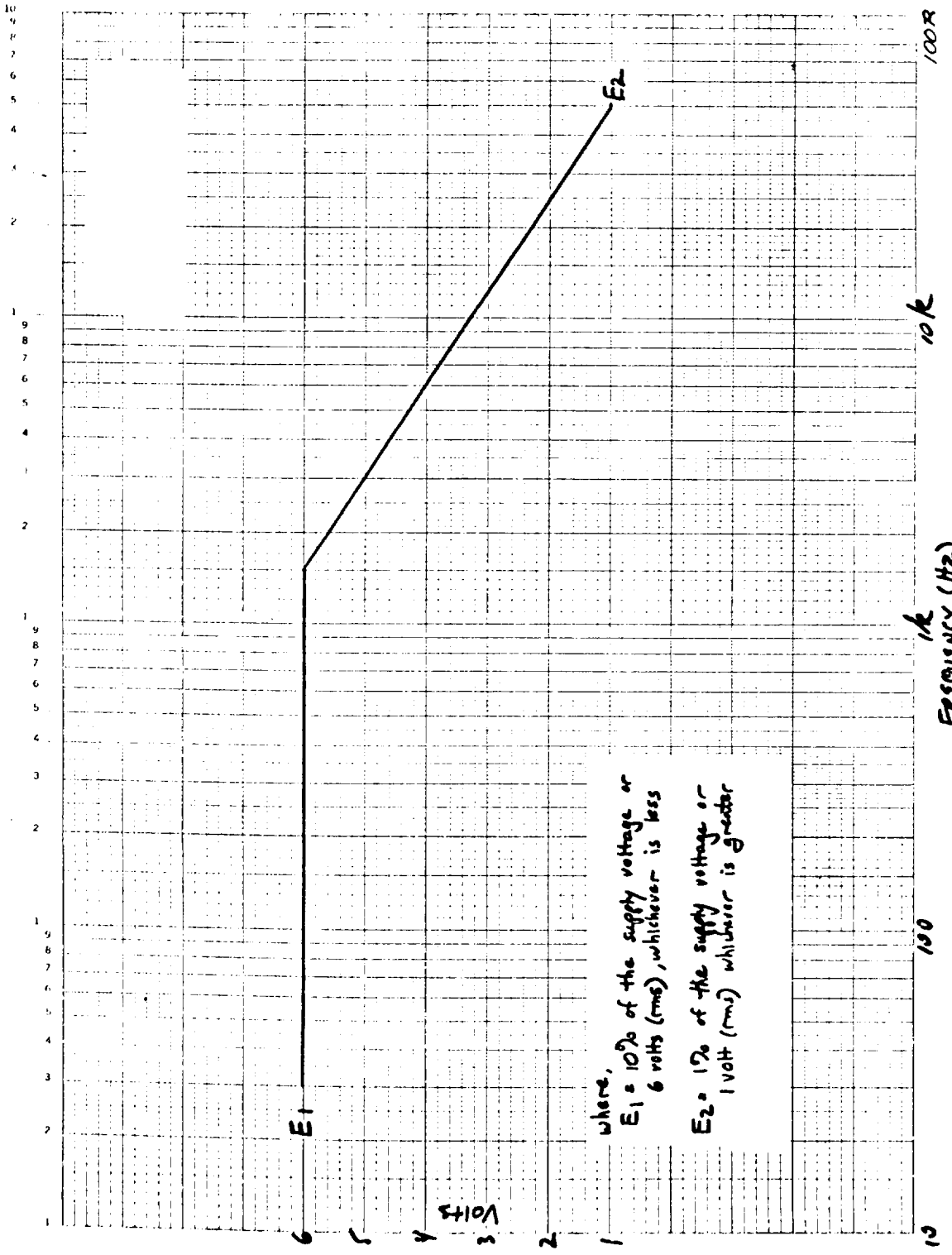


FIGURE 7. Limit for CS01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

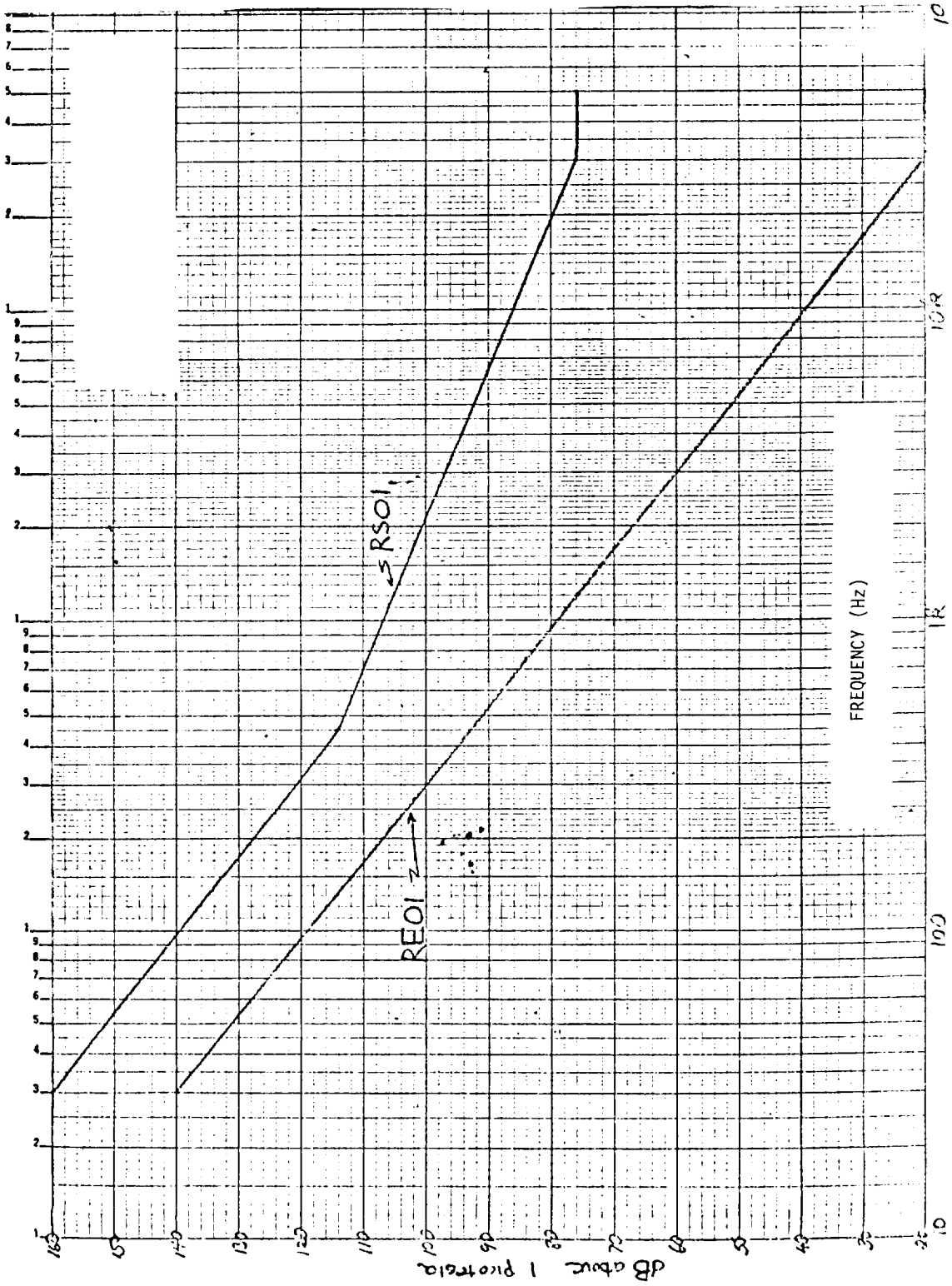


FIGURE 8. Limits for RE01 and RS01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.14.3 Overload protection. Electrical short circuit and overload protection is required. The protective device shall be mounted on the front panel and shall be provided by means of a circuit breaker with an indicator to show when the circuit breaker has opened the circuit(s). (The power indicator may be used to indicate an open circuit breaker by being off.) An overload trip unit of the circuit breaker shall be connected in series with each ungrounded conductor as determined by the National Electrical Code NFPA 70-1978 for grounded source power and shall be located on the load side of the power switch.

3.2.14.4 Panel illumination. Front panel illumination is not required.

3.2.14.5 Interlock switch. An interlock switch shall be provided to remove all electrical power from the equipment when the front panel is opened. This switch shall have a defeat position for maintenance which is disengaged when the front panel is closed.

3.2.15 I/O connectors. I/O connectors shall be provided on the front panel and also at the rear of the unit. The front panel connectors are test points and the rear connectors are the system interface connectors. The front and rear panel connectors shall be Type BNC as specified in MIL-C-39012, 3.2.15.1 and 3.2.15.2.

3.2.15.1 Front panel connectors. Front panel connectors shall be as follows:

		Quantity
a.	Auxiliary/internal trigger out	1
b.	IFF trigger in	1
c.	IFF trigger out	1
d.	Radar A pre-trigger in	1
e.	IFF video in	1
f.	Retimed IFF video out	1
g.	Gated IFF video out	1
h.	Retimed mode tags out	1
i.	Mode tags out	1
j.	Radar B trigger in	1

3.2.15.2 Rear panel connectors. Rear panel connectors shall be as follows:

		Quantity
a.	Auxiliary trigger in	1
b.	Auxiliary/internal trigger out	1
c.	M4 trigger out	2
d.	Selective identification feature (SIF) trigger out	2
e.	Display trigger out number 1	1
f.	Display trigger out number 2	1
g.	IFF trigger in	1
h.	IFF trigger out	1
i.	Radar A pre-trigger in	1
j.	Radar A pre-trigger out	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

	Quantity
k. Radar A end of sweep (EOS) trigger in	1
l. Radar A EOS trigger out	1
m. Radar B trigger in	1
n. Radar B trigger out	1
o. IFF video in	1
p. Retimed IFF video out	3
q. Gated IFF video out	3
r. Mode tags in	1
s. Retimed mode tags out	3
t. Gated mode tags out	3
u. Selectable trigger out number 1	1
v. Selectable trigger out number 2	1
w. Selectable trigger out number 3	1
x. 115 VAC power and ground	1

3.2.16 I/O characteristics. I/O characteristics shall be as follows:

a. Auxiliary trigger in. Positive pulse 150 to 3000 pulse-per-second (pps); 0.3 µs to 20 µs pulsewidth with rise time up to 25 percent of the duration of 0.04 µs per V times the amplitude and fall time of up to 1 to 5 times the rise time; 1.5 V to 55 V amplitude. A 75-ohm termination shall be provided. This trigger shall be used as the external trigger source to produce the reference pulse train. Solderless links shall be provided to internally select either the delayed IFF trigger (see 3.2.16f) or the Radar A pseudo pre-trigger (see 3.2.17.2.2b).

b. Auxiliary/internal trigger out. Twenty ±5 V into 75-ohm load with same pulse characteristics as specified in 3.2.16a. This is the buffered output of the auxiliary input trigger or the internal trigger source selected by the trigger source switch.

c. Mode 4 trigger out. Positive pulse, 1 ±0.5 µs width; 0.02 µs to 0.1 µs rise time and fall time; amplitude, 20 ±5 V into a 75-ohm load. This trigger is derived from the trigger generator function of the equipment, and shall be used for Mode 4 trigger in the independent mode operation. In the auxiliary trigger operation, it may be used as a buffered undelayed and counted down trigger.

d. SIF trigger out. Same pulse characteristics as specified in 3.2.16c, but adjustable delay of 50 µs to 500 µs from Mode 4 trigger out. In the independent mode operation, it shall be used as a SIF trigger to the interrogator. In the auxiliary trigger operation, it may be used as a delayed and counted down trigger.

e. Display trigger out. Same pulse and delay characteristics as specified in 3.2.16d. In the independent mode operation, this trigger may be used as zero or display trigger. In the auxiliary trigger operation, it may be used as a delayed trigger.

f. IFF trigger in. Positive pulse 0.3 µs to 5.0 µs pulsewidth; 0.02 µs to 0.1 µs rise and fall time; amplitude, 1.5 V to 55 V. A 75-ohm termination shall be provided. When used, this trigger normally comes from the interrogator or another equipment to initiate the storage (write) cycle. An adjustable 0 µs to 100 µs delay (with delay jitter less than ±0.1 µs) shall be provided before this trigger is routed to the timing and control function for initiating the storage cycle or to the auxiliary trigger input (see 3.2.16a).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

g. IFF trigger out. Same characteristics as specified in 3.2.16c. This buffered trigger shall be used to provide the same storage cycle timing when another equipment is used. Solderless links shall be provided to select either the external IFF input trigger (delayed or undelayed) (see 3.2.16f) or the timing and control function trigger that initiates the storage cycle (see 3.2.17.1.1d).

h. Radar A pre-trigger in. Same characteristics as specified in 3.2.16a. A 75-ohm termination shall be provided. This signal comes from the independently triggered Radar A and is used to initiate the recall (read) cycle of stored IFF video and mode tags in the equipment.

i. Radar A pre-trigger out. Same characteristics as specified in 3.2.16h. This is the buffered output of the Radar A input trigger.

j. Radar A EOS trigger in. Same characteristics as specified in 3.2.16a. A 75-ohm termination shall be provided. When used, this trigger shall terminate the IFF video output from storage.

k. Radar A EOS trigger out. Same characteristics as specified in 3.2.16j. This is a buffered output of the Radar A EOS input trigger.

l. Radar B trigger in. Same characteristics as specified in 3.2.16a. A 75-ohm termination shall be provided. These pulses come from the slaving radar that is triggering the interrogator and shall be used to terminate the IFF video output from the video gating function.

m. Radar B trigger out. Same characteristics as specified in 3.2.16.l. This trigger is a buffered output of the Radar B input trigger.

n. IFF video in. Positive 1.5 V to 3 V amplitude. A 75-ohm termination shall be provided. This signal comes from the interrogator and shall be read into the storage of the equipment.

o. Retimed video out. Positive pulses adjustable 1.5 V to 3 V terminated into 75-ohm load. This is the retimed video read-out of storage which is time-synchronized to Radar A trigger.

p. Gated IFF video out. Positive adjustable 1.5 V to 3 V terminated into 75-ohm load. This is the input IFF video gated to be range compatible with Radar B target video.

q. Mode tags in. Positive pulse train 2 V to 20 V amplitude. A 75-ohm termination shall be provided. This signal comes from the interrogator and shall be read into storage in the equipment.

r. Retimed mode tags out. Positive pulses adjustable 2 V to 20 V into 75-ohm load. This is retimed mode tags read-out of storage which is time-synchronized to Radar A trigger.

s. Mode tags out. Positive pulses adjustable 2 V to 20 V into 75-ohm load. This is the buffered mode tags signal from interrogator.

t. 115 VAC in. 115 V $\pm$10 percent, 57 Hz to 63 Hz and 360 Hz to 440 Hz single-phase.

u. Selectable trigger out. Positive pulses 20 $\pm$5 V into a 75-ohm load, and a 0.02 µs to 0.1 µs rise time and fall time. The pulse repetition rate and width shall be that of the selected trigger.

3.2.16.1 Input termination. All input terminations shall be electrically and physically removable at the rear panel connector. Unsoldering is an acceptable method of removal of the terminations.

3.2.16.2 Direct current (DC) isolation. The video signal inputs shall be capacitively coupled to ensure proper system operation with signals which have a DC reference other than ground.

3.2.16.3 Output reference level. Except as hereinafter specified, the video signal outputs of all equipments produced in accordance with this specification shall be referenced to ground -0.1 V +0.2 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.16.4 Input and output protection. All trigger and video outputs specified in 3.2.15.1 and 3.2.15.2 shall withstand either a continuous and intermittent open circuit condition or a continuous and intermittent short circuit condition without any damage to the internal circuit of the equipment.

3.2.16.5 Input thresholding. A fixed input threshold shall be provided for all input triggers and video input signals (IFF and mode tags) such that trigger and video input signals below the level of 1.2 V shall not be processed by the equipment.

3.2.17 Functional capabilities. The functional capabilities specified herein are requirements of the equipment. FIGURE 9 is provided as a guide to show the interrelationship of the required functional capabilities.

3.2.17.1 Trigger generator. The trigger generator function shall provide the necessary timing triggers for the equipment and the interrogator. It shall be possible for the trigger generator to operate independently (free-running) or slaved to an external incoming auxiliary trigger. An independent-auxiliary trigger source switch shall provide the trigger source selection of the trigger generator function.

3.2.17.1.1 Independent mode operation. When the equipment is operating in the independent mode, the trigger generator shall provide reference pulses with a PRF that is adjustable internally from 100 pps to 410 pps in one pps increments. Provisions (via solderless link settings) shall be made to extend PRF range to 1000 pps in one pps increments. From these reference pulses the following delayed triggers (with delay jitter less than $\pm$0.1 µs), adjustable in increments less than 1.3 µs shall be provided:

a. Two separately buffered Mode 4 triggers to the output connectors. These Mode 4 triggers shall have the same PRF as the referenced pulses and zero delay (nominal).

b. Two separately buffered SIF triggers to the output connectors. The two SIF triggers shall have the same PRF as the reference pulses; and via a single adjustment control, these SIF triggers shall have identical delays that are adjustable from 50 µs to 500 µs from the reference pulses.

c. Two separately buffered display triggers to the output connectors. The two display triggers shall have the same PRF as the reference pulses and two separately adjustable delays that are internally adjustable from 50 µs to 500 µs from the reference pulses. The adjustable delays of the two delayed triggers shall be independent of the delay of the SIF trigger specified in 3.2.17.1.1b.

d. An internal IFF trigger to the timing and control function of the equipment. This trigger to the timing and control function shall have the same PRF as the reference pulses and a delay that is internally and independently adjustable from 50 µs to 500 µs from the reference pulses. This trigger is used by the timing and control to initiate the write command to the digital storage unit of the equipment.

3.2.17.1.2 Auxiliary mode operation. When the trigger generator function is operating in the auxiliary mode, the auxiliary trigger shall replace the internal trigger source from which the reference pulses and the seven delay triggers identical to those specified in 3.2.17.1.1 shall be generated. The PRF of the reference pulses generated shall be governed by a countdown circuit (which limits the maximum PRF of the reference pulses) such that all auxiliary trigger rates up to the countdown rate shall be passed without being counted down and all auxiliary trigger rates greater than the countdown rate shall be counted down at the largest sub-multiple below the countdown rate. The auxiliary trigger rate at which countdown begins shall be made by the same control (and have the same adjustment ranges) as used in the independent mode specified in 3.2.17.1.1.

3.2.17.1.3 PRF and countdown control. When the equipment mode switch is changed from independent to auxiliary or vice versa, the PRF in the independent mode of operation and the countdown rate in the auxiliary mode of operation for the reference pulses shall be the same. No changes in the PRF and countdown adjustment control or the adjustment ranges shall be required to achieve the same PRF (in the independent mode) and countdown rate (in the auxiliary mode). The countdown circuit shall be capable of being disabled and enabled and shall be independent of the PRF and countdown control and range settings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

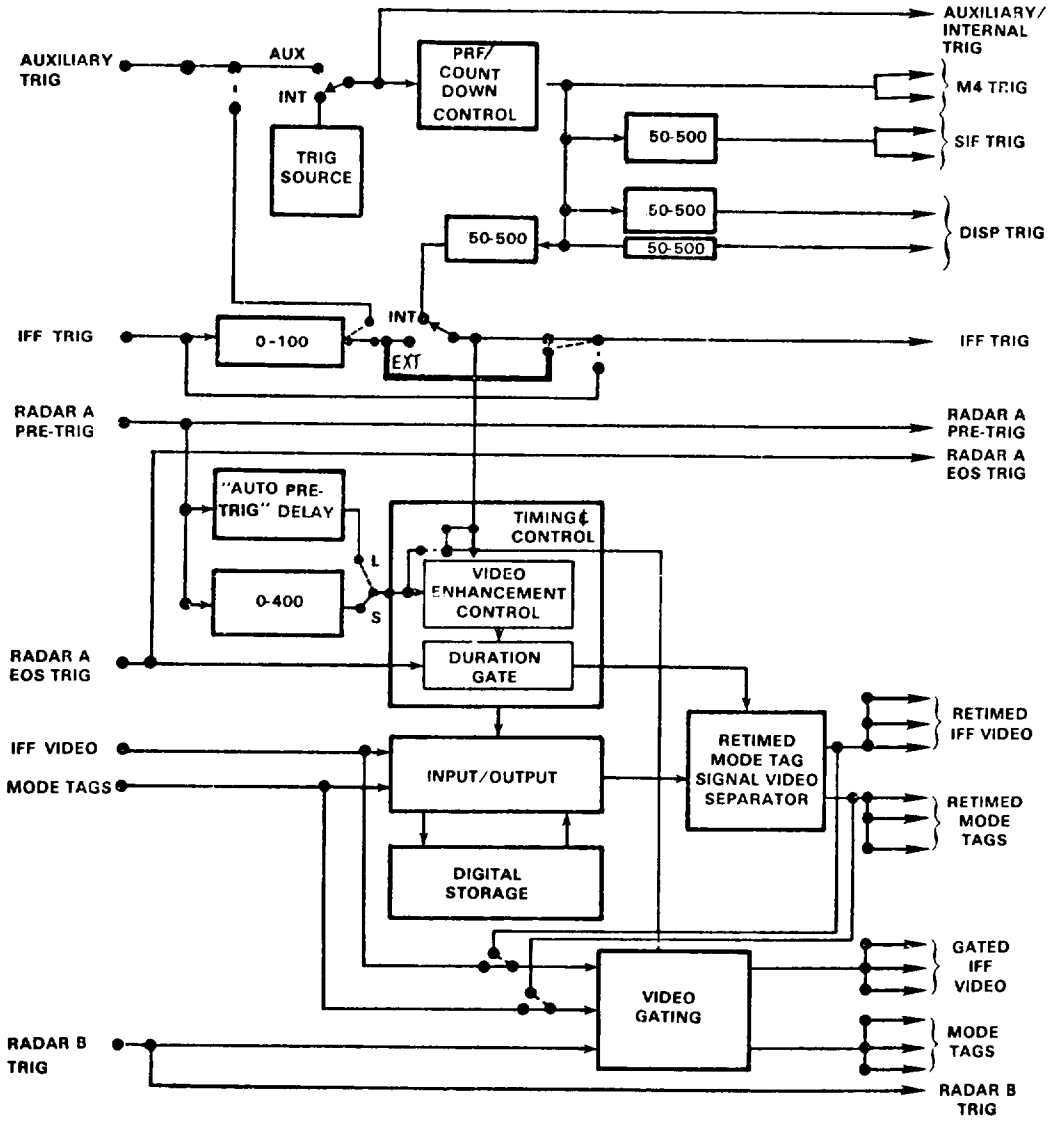


FIGURE 9. Equipment functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.17.1.4 Simultaneous selection of triggers. The auxiliary trigger or the triggers generated by the internal trigger generator and the IFF trigger input (see 3.2.16f) shall be selectable simultaneously. When such a selection is made, the auxiliary trigger or the triggers generated by the internal trigger generator shall control the delay triggers (see 3.2.17.1.1), and the IFF trigger input shall initiate the IFF video recall interval and storage cycle (see 3.2.17.2.1).

3.2.17.2 Timing and control. The timing and control shall provide the timing and gating functions of the equipment. Provisions via a switch shall be made such that IFF trigger specified in 3.2.16f and 3.2.17.1.1d are selectable. The timing and control function shall:

a. Command the I/O function to gate mode tags and IFF video into and out of the digital storage function

b. Command the mode tag-signal video separator function to gate retimed mode tags and IFF video from the digital storage function, via the I/O function to the output buffers

c. Command the video gating function to gate the input IFF video to the output buffers

3.2.17.2.1 Mode tag and IFF video storage. The IFF trigger selected herein shall initiate the IFF video recall interval and the storage cycle of mode tags and IFF video. The IFF video recall interval shall be adjustable from 500 µs to 10,000 µs, in increments of less than 10.3 µs. The mode tags and IFF video shall be placed into storage within 1 µs after receipt of the IFF trigger by the timing and control function. The storage cycle shall terminate when maximum storage capacity has been reached or when a second IFF trigger has been received.

3.2.17.2.2 Mode tag and IFF video termination and recall. Via solderless links, two selectable radar delays (with delay jitter less than 200 nanoseconds (ns) shall be provided before the incoming Radar A pre-trigger is routed to the timing and control function, for recall initiation, as follows:

a. An adjustable delay of the incoming Radar A pre-trigger from within one storage clock cycle to 400 µs in increments of less than 1.3 µs.

b. An adjustable delay of the incoming Radar A pre-trigger, having a stable (nonjittered) PRF, to produce a pseudo pre-trigger (reference to incoming Radar A pre-trigger timing) that is adjustable from 0 µs to 500 µs, in increments of less than 1.3 µs, prior to the next incoming Radar A pre-trigger. This radar delay shall automatically compensate for PRF changes in the incoming Radar A pre-trigger such that the pseudo pre-trigger to the timing and control function shall be correct within two pulse repetition times (PRT) or less, after the new Radar A pre-trigger PRF has stabilized. Only one of these radar delays shall be selected at a time. Termination and recall of the retimed mode tags and IFF video shall be as specified in 3.2.17.2.2.1 through 3.2.17.2.2.2.2.

3.2.17.2.2.1 Mode tags and IFF video recall cycle termination. Termination of the retimed mode tags and IFF video recall cycle shall be accomplished within 640 ns by the means specified in 3.2.17.2.2.1.1 and 3.2.17.2.2.1.2.

3.2.17.2.2.1.1 Automatic termination. Unless other means of termination are selected as specified herein, the retimed mode tags and IFF video recall shall automatically be terminated when maximum storage capacity of retimed IFF video has been outputted.

3.2.17.2.2.1.2 Selectable termination. By means of solderless links, one or any combination of the following shall be selectable to terminate the mode tags and IFF video recall cycle:

a. Radar A EOS trigger

b. When a manually adjustable output duration gate has terminated. With respect to the time when the initial retimed mode tags and IFF video is first recalled from storage this manually adjustable output duration gate shall be adjustable from 100 µs to the maximum storage capacity in increments of 10.3 µs.

c. Upon receipt of a Radar A pre-trigger (plus selected radar delay)

The recall cycle shall terminate upon first occurrence of the selected means of termination specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.17.2.2.2 Mode tag and IFF video recall cycle initiation. Each IFF video recall cycle, which recalls and sends the most recently stored mode tags and IFF video from storage to the proper outputs in the same time sequence of the originally stored video, shall be initiated as specified herein, selectable by solderless links. In addition, an advanced video range recall provision, via solder link selections, shall be made to recall IFF video such that after mode tags have been recalled, the recalled IFF video shall begin at a preselected range that is beyond the originally stored zero range. However, the output of the preselected range recalled IFF video shall begin within one-twelfth of a nautical mile after the mode tags (originally stored range zero). The preselected range shall be adjustable from 0 to 10 nautical miles (continuously or in increments no greater than one-twelfth of a nautical mile) beyond the originally stored zero range.

3.2.17.2.2.2.1 Normal recall initiation. If the Radar A pre-trigger has been selected for termination of mode tags and IFF video in accordance with 3.2.17.2.2.1.2, then the stored mode tags and IFF video shall be recalled from digital storage within 1 s after receipt of the Radar A pre-trigger (plus the selected radar delay). However, if the Radar A pre-trigger has not been selected to terminate the previous recall cycle in accordance with 3.2.17.2.2.1.2, then any additional recall(s) by Radar A pre-trigger(s) shall be inhibited until the previous recall cycle has been terminated automatically (see 3.2.17.2.2.1.1) or by one of the other terminating methods specified in 3.2.17.2.2.1.2. Once termination has occurred, the next Radar A pre-trigger (plus the selected radar delay) shall initiate a new mode tag and IFF video recall cycle. Radar A pre-triggers, which occur after an IFF trigger, and within the preselected advanced range time, shall not output retimed mode tags and IFF video when so selected by solderless links.

3.2.17.2.2.2.2 Alternate recall initiation. An alternate method for separate recall of mode tags and IFF video shall provide, by solderless links, the means of selecting only one of the following triggers: a) the external IFF trigger, b) the auxiliary trigger, c) the Radar A EOS trigger, or d) a trigger occurring at the termination time of the manually adjustable output duration gate, to terminate the previous recall cycle and start a new recall cycle which separately recalls the most recently stored mode tags and IFF video from storage. The stored IFF video associated with the mode tags shall be recalled only upon receipt of the Radar A pre-trigger. In addition, termination of the previous IFF video recall cycle shall also be selectable in any combination of methods specified in 3.2.17.2.2.1.1. However, if the Radar A pre-trigger has not been selected to terminate the previous recall cycle in accordance with 3.2.17.2.2.1.1, further IFF video shall not be recalled by any other Radar A pre-trigger until IFF video recall has been terminated by the selected method(s) specified in 3.2.17.2.2.1 or the methods specified herein for previous recall cycle termination and separate recall of new mode tags has occurred. Provision shall be made by selectable links so that when an IFF trigger occurs before Radar A pre-trigger and after the selected trigger which initiates mode tag recall, the IFF trigger shall not initiate a new storage cycle.

3.2.17.2.3 Video enhancement. The mode tags and IFF video shall be recalled from storage by Radar A pre-trigger only if Radar A pre-trigger occurs within the IFF video recall interval initiated by the receipt of the IFF trigger. Radar A pre-trigger(s) occurring after the video recall interval shall not be capable of recalling mode tags and IFF video from storage until after the receipt of an IFF trigger. If more than one Radar A pre-trigger occurs within the video recall interval, the IFF video output by the previous Radar A pre-trigger shall terminate just immediately before initiation of the new recall cycle to ensure that no second time around video signal is outputted. Provisions via switches shall also be made to enable or disable an enhancement counter (presettable from one to seven) which determines the number of IFF video recall cycles that can be executed per IFF video storage cycle. When the enhancement counter is utilized (selected), the methods for terminating the retimed IFF video output of the last executed recall cycle shall be either as specified in 3.2.17.2.2, or by means of selection by solder-less links, the output of the manually adjustable duration gate and only the output of this gate shall terminate the last executed recall cycle regardless of the method(s) selected from 3.2.17.2.2 for termination of the recall cycle. If more than one IFF trigger occurs before the next Radar A pre-trigger occurrence only the most recent IFF trigger stored mode tags and IFF video shall be recalled.

3.2.17.2.4 Storage and recall cycle overlapping. Regardless of synchronous or asynchronous IFF and Radar A triggers, the storage cycle and recall cycle operations shall be capable of overlapping.

3.2.17.2.5 Total jitter requirement. When any given mode tags and IFF video are recalled by a series of input Radar A pre-triggers (that is, PRT to PRT), then the total jitter of the recalled mode tags and IFF video (measured at the retimed IFF video and retimed mode tag outputs) with respect to the input Radar A pre-trigger shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

a. No more than 120 ns when the Radar A pre-trigger is routed to the timing and control function in accordance with 3.2.17.2.2a.

b. No more than 200 ns when the Radar A pre-trigger is routed to the timing and control function in accordance with 3.2.17.2.2b.

The total jitter specified herein shall include jitter from all sources including, but not limited to: storage, recall, data, and delay. Also, the total jitter requirements specified herein are applicable over a reduced operating temperature range of $0^{\circ}C$ to $50^{\circ}C$ (see 3.2.5.1). An additional 80 ns of jitter, as specified herein, is allowed over the operating temperature ranges of $-28^{\circ}C$ to $0^{\circ}C$ and $+50^{\circ}C$ to $+65^{\circ}C$.

3.2.17.2.5.1 Jitter - alternate mode tag recall. When any given mode tags are recalled by a series of initiating triggers other than the Radar A pre-trigger (see 3.2.17.2.2.2.2), then the total jitter of the recalled mode tags (measured at the retimed mode tag outputs) with respect to the initiating triggers shall be not more than 1.0 s. This total jitter shall include jitter from all sources including, but not limited to: storage, recall, data, and delay.

3.2.17.3 Clock function. The clock function shall provide, as a minimum, the timing pulses to clock mode tags and IFF video into and out of the digital storage unit. The clock shall have a frequency such that a storage time resolution of at least 0.1 µs is obtained. A resolution of 0.08 µs or greater shall be the design goal. The stability shall be better than ±0.0025 percent.

3.2.17.4 Digital storage. The digital storage shall retain mode tags and IFF video up to a minimum of 300 nautical miles upon command from the timing and control function. The storage device may be either a random access memory or a shift register memory capable of holding the stored video signal for a period as short as one clock cycle to longer than 10 milliseconds (ms) The digital storage shall be capable of overlapping read and write (storage and recall) operations. The read and write operation when overlapping, may be simultaneous, but shall not differ in time by more than one clock cycle. The stored signal resolution and the output signal resolution shall be such that the signal pulsewidths and spacings within an IFF video reply signal shall not differ by greater than 0.1 µs from the input signal for each pulse of the reply. Accumulation of tolerance beyond 0.1 µs shall not be acceptable.

3.2.17.4.1 Output signal timing. Upon command from the timing and control function, the stored mode tags and IFF video signals shall be routed to the output buffers via the I/O function in the same timing and sequence as the original input IFF video signals including mode tags (see 3.2.17.2.1 and 3.2.17.2.2). The output signal timing of the digital storage and associated circuits shall be such that target range accuracy is not degraded by more than one-twelfth of a nautical mile throughout the range of 300 nautical miles.

3.2.17.5 I/O. The I/O function shall direct incoming mode tags and IFF video into storage and from storage to the output buffers upon command from the timing and control function. The input IFF video signals from the interrogator shall be positive pulses of 1.5 V to 3 V amplitude when terminated with 75 ohms. The input mode tags from the interrogator shall be positive pulses of 2 V to 20 V amplitude when terminated with 75 ohms.

3.2.17.6 Mode tag-signal video separator. The mode tag-signal video separator function shall accept the stored (retimed) mode tags and IFF video signal from digital storage via the I/O function, separate the mode tags from the IFF video signal, and direct the separated mode tags and IFF video to the proper output buffers. Solderless links shall be provided to permit the separation of mode tags and IFF video to separate outputs, or combined such that both mode tags and IFF video are outputted in serial to the retimed video outputs. Mode tags shall be received from the interrogator in advance of IFF video and as such shall be stored first. There shall be an adjustable time from the IFF trigger, specified in 3.2.16f and 3.2.17.1.1d, recognized as the mode tag interval. The mode tag interval shall be adjustable from 1 µs to 160 µs in less than 1.3 µs increments. This interval shall not jitter by more than 200 µs with respect to the IFF trigger.

3.2.17.7 Video gating. The video gating function shall provide the time gate through which the incoming mode tags and IFF video from the interrogator shall pass, with a propagation delay of no more than 200 ns, from the input connector to separate video output connectors, via output buffers. The video gating function shall also be capable of selecting either the mode tags and IFF video inputs from the interrogator or the retimed mode tags and IFF video output from the retimed mode tag-signal video separator function. The time gate shall prevent second time around IFF target video from appearing on the gated IFF video output of the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.2.17.7.1 Video gate function. The video gating function via solderless links shall accept the IFF gating trigger (from the timing and control function), the mode tags and IFF video from the external interrogator, or the retimed mode tag-signal video separator, and the radar trigger from Radar B. Via solderless links, the IFF gating trigger shall be selectable to occur a) at the same rate and timing as the IFF trigger, b) at the same rate and timing as the Radar A pre-trigger to the timing and control function, and c) at the rate of once following each IFF trigger and at the Radar A pre-trigger timing to the timing and control function. Upon receipt of the IFF gating trigger, the video gating function shall output the incoming mode tags and IFF video for a variable period starting at the time of IFF gating trigger occurrence and ending with the arrival of Radar B trigger. In addition, an adjustable, internal video time gate trigger occurring 100 μs to 4000 μs after the IFF gating trigger shall be provided to work in conjunction with or without Radar B trigger to end the gated IFF video outputted. This internal video time gate trigger shall be enabled or disabled via solderless links. The internal video time gate trigger shall be adjustable in increments of less than, or equal to, 10.3 μs.

3.2.17.7.2 Video gate disabling. The video gating function shall be capable of being disabled to provide buffered outputs of the entire selected incoming IFF video.

3.2.17.8 Trigger output buffers. Fourteen (total) output buffers shall be provided for triggers. Each buffer shall have pulse amplitude of 20 ±5 V when terminated into a 75-ohm load. The buffered outputs shall be protected against continuous and intermittent direct short or open circuiting. One output buffer shall be dedicated for each of the following triggers: a, e, f, g, and h, and two output buffers shall be dedicated for triggers b, c, and d. Via multiposition switches, each of the remaining three buffers shall be capable of selecting any of the following triggers:

a. Auxiliary/internal trigger. One buffer is normally connected

b. Mode 4 trigger. Two buffers are normally connected

c. SIF trigger. Two buffers are normally connected

d. Display trigger. Two buffers are normally connected

e. IFF trigger. One buffer is normally connected

f. Radar A pre-trigger. One buffer is normally connected

g. Radar A EOS trigger. One buffer is normally connected

h. Radar B trigger. One buffer is normally connected

There are two display triggers referenced in d; each one is routed to a single buffer for a total of two. The selectable trigger function shall select display trigger number 1 (see 3.2.15.2u).

3.2.17.9 IFF video output buffers. The following six (total) output buffers shall be provided in IFF video. The buffers shall have adjustable output pulse amplitude from 1.5 V to 3 V when terminated into a 75-ohm load. The buffered outputs shall be protected against continuous and intermittent direct short circuiting.

a. Retimed IFF video. Three output buffers shall be provided for retimed IFF video coming from the digital storage function via the I/O and mode tag signal video separator functions.

b. Gated IFF video. Three output buffers shall be provided for gated IFF video coming from the video gating function.

3.2.17.10 Mode tags output buffer. The following six (total) output buffers shall be provided for IFF mode tags. The buffers shall have adjustable output pulse amplitude from 2 V to 20 V when terminated into a 75-ohm load. The buffers shall be protected against continuous and intermittent direct short circuiting.

a. Retimed mode tags. Three output buffers shall be provided for retimed mode tags coming from the digital storage via the I/O and mode tag-signal video separator functions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

b. Mode tags. Three output buffers shall be provided for the inputted mode tags coming from the video gating function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to perscribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

1. Production inspection (Group A) (see 4.4.1)

2. Production control inspection (Group B) (see 4.4.2)

3. Environmental inspection (Group C) (see 4.4.3)

c. Reliability qualification testing (see 4.6)

d. Maintainability demonstration (see 4.7)

e. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each equipment offered for delivery. The inspection shall comprise such examination and tests which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, test of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests specified in TABLE II (Group A).

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. The inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examination and tests specified in TABLE II (Group B).

4.4.2.1 Sampling for Group B inspection. Equipment submitted for Group B sampling tests shall be selected by the contractor under the supervision of the Quality Assurance Representative (QAR) and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level (AQL) shall be 6.5 percent defective and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1, the inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as, extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examination and tests specified in TABLE II (Group C).

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The equipment shall then be retested.

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to and have passed production control or environmental inspections, or both, may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall conform to the following quality assurance provisions of MIL-E-16400:

a. Test facilities, chambers, and apparatus

b. Standard test conditions

c. Performance of test

4.5 Test methods.

4.5.1 Examinations and tests. Unless otherwise specified herein, the inspections specified in TABLE II shall be performed in accordance with the test methods specified in MIL-E-16400.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

TABLE II. Examination and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination:						
Weight	3.2.6.4	4.5.1	X	X		
Size	3.2.6.3	4.5.1	X	X		
Parts and materials	3.2.3.1, 3.2.3.3, 3.2.4, 3.2.6, 3.2.13.1, 3.2.14, 3.2.15	4.5.1	X	X		
Finish	3.1	4.5.1	X	X		
Marking	3.1	4.5.1	X	X		
Safety	3.1	4.5.1	X	X		
General workmanship	3.1	4.5.1	X	X		
Workmanship screen	3.2.11.1, 3.2.11.2	4.5.2		X		
Performance	3.2.16, 3.2.17	4.5.1	X	X		
Low temperature	3.2.5.1	4.5.1	X			X
High temperature	3.2.5.1	4.5.1	X			X
Humidity	3.1	4.5.1	X			X
Salt fog (spray)	3.2.5.2	4.5.1	X			
Shock	3.1	4.5.1	X			X
Vibration	3.2.5.3	4.5.1	X			X
Inclination	3.2.5.4	4.5.1	X			X
Magnetic field environment	3.2.5.5	4.5.1	X			X
Enclosure	3.2.6.6	4.5.1	X		X	
Steady-state voltage and frequency	3.2.13	4.5.1	X		X	
Transient voltage	3.2.13	4.5.1	X			
Transient frequency	3.2.13	4.5.1	X			
Spike voltage	3.2.13	4.5.1	X			
Power interruption	3.2.13	4.5.1	X		X	
Power and power factor	3.2.13	4.5.1	X		X	
Leakage current	3.1	4.5.4	X			
EMC	3.2.12	4.5.5	X			X
Thermal design	3.1	4.5.1	X			
Interchangeability	3.2.3.2	4.5.3	X			
Reliability test and demonstration	3.2.8	4.6	X			
Maintainability demonstration	3.2.7	4.7	X			
Power supply	3.2.13.2	4.5.6	X			

4.5.2 Workmanship screen. The workmanship screen shall consist of random vibration and temperature cycling. The screening may be performed at the module, drawer, or end item level. The random vibration shall be performed on each equipment, including cables and connectors, prior to conducting temperature cycling test (see 4.5.2.2).

4.5.2.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and shall be subjected to the vibration conditions shown in FIGURE 10. The control accelerometer shall be located next to one of the mounting points of the item under test. The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz and a 100 Hz bandwidth, above 500 Hz. The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. Any failure occurring during test shall require corrective action before the test is resumed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

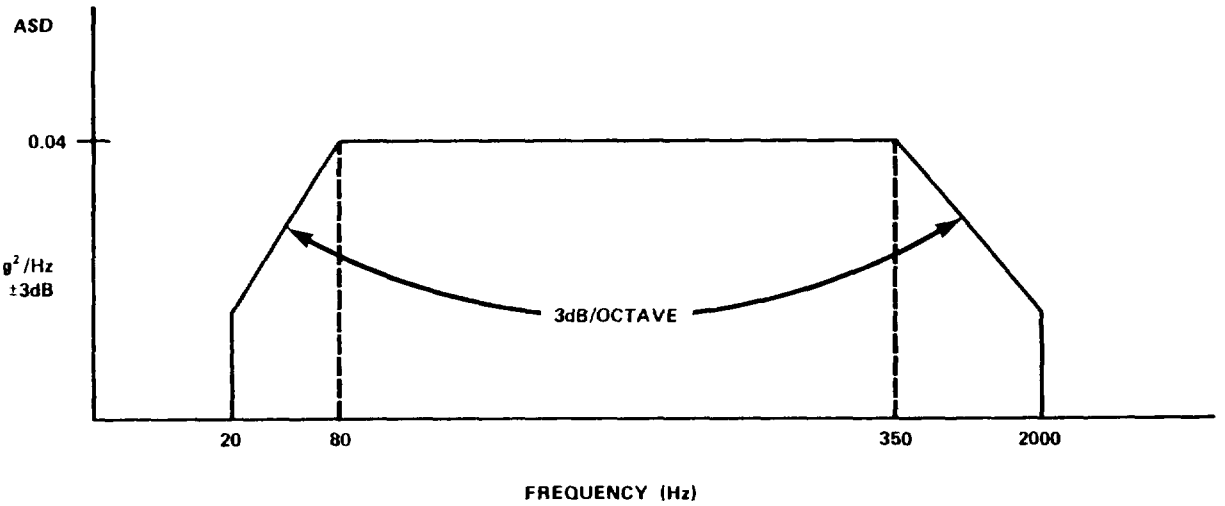


FIGURE 10. Random vibration curve.

4.5.2.2 Temperature cycling. Each equipment shall be subjected to the temperature curve shown in FIGURE 11. the number of cycles required shall be determined by the electrical and electronic parts count and the parts count shown in TABLE III. The temperature rate of change shall be not less that 5°C per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition.

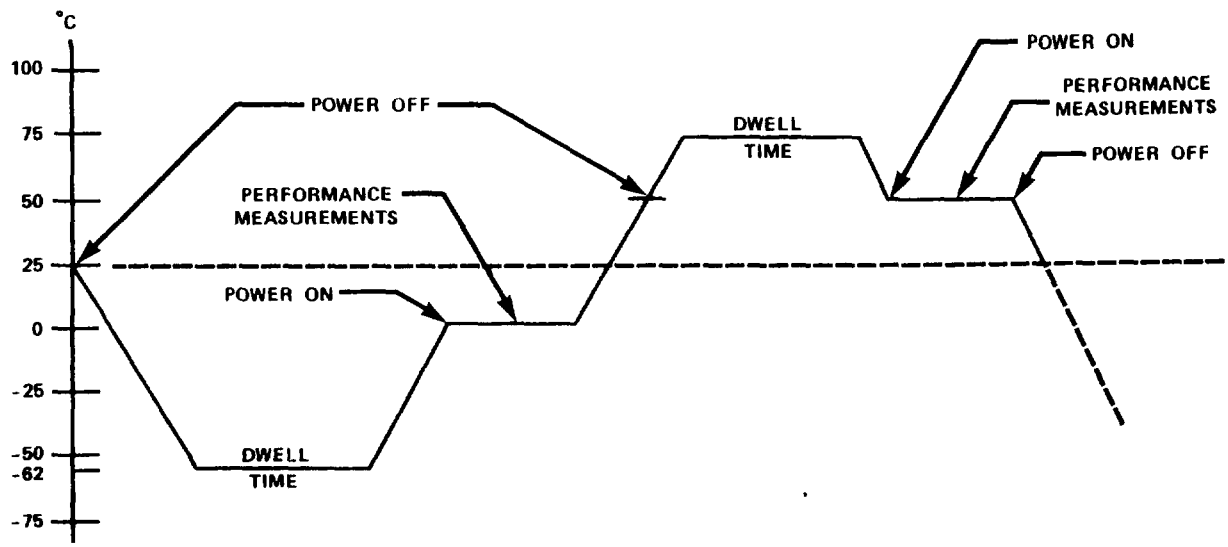


FIGURE 11. Temperature curve (one cycle).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

TABLE III. Electrical and electronic parts count.

Electrical and electronic parts count	Number of temperature cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

4.5.3 Interchangeability demonstration. The equipment shall be examined and tested to verify and demonstrate compliance with 3.2.3.2 by utilizing at least two equipments and interchanging each interchangeable assembly, subassembly, module, and so forth between the two equipments. Satisfactory equipment performance shall be demonstrated prior to and after the interchange of the interchangeable assemblies, subassemblies, modules, and so forth. Interchangeability of piece parts shall be verified by examination.

4.5.4 Leakage current test. Leakage current shall be measured at 115 V, 60 Hz.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.4.1 Equipment test connections. After power removal, each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment and the source power shall be opened during the test. Observe warning statement. The equipment shall be connected as shown in FIGURE 12.

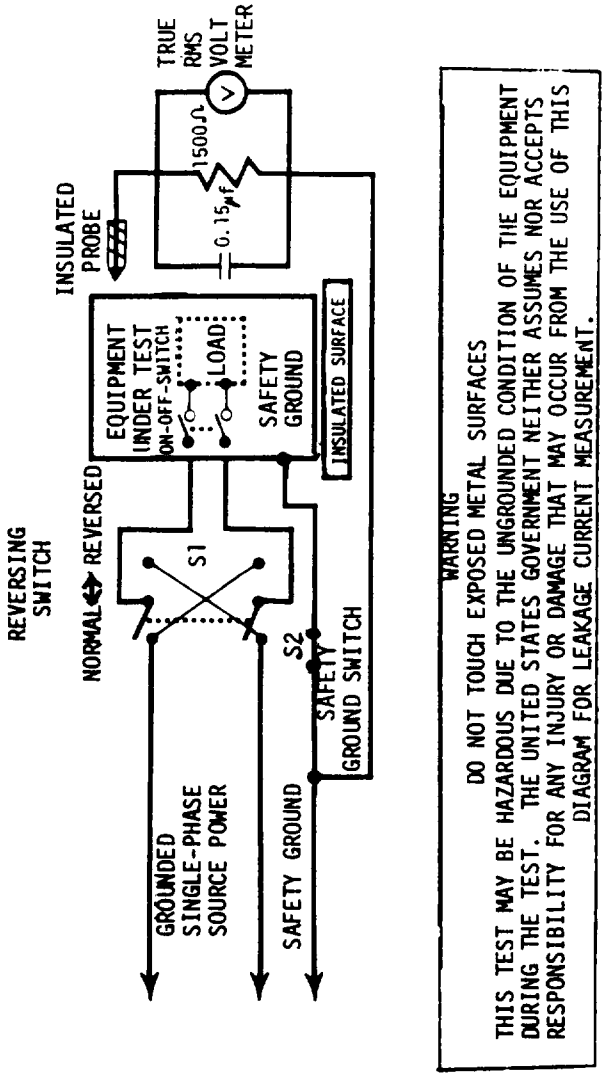
4.5.4.2 Measurement. Leakage current shall be measured on equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 milliamperes (mA) of leakage current. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 12. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.5 EMC test. The requirements of 3.2.12 shall be verified by tests performed in accordance with MIL-STD-462. For environmental production tests (Group C of TABLE II only) tests CE03 and RE02 shall be performed.

4.5.6 Power supply tests. Power supply tests shall be conducted to determine conformance to 3.2.13.2. These tests may be performed with the power supply external to the equipment.

4.6 Reliability qualification testing. Reliability qualification testing shall be conducted in accordance with Test Plan VIIIC of MIL-STD-781 for Category 3A equipment with a minimum of two equipments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.



GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING. S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED.
7. ON-OFF SW ON. REPEAT STEP 4
8. ON-OFF SW OFF. REPEAT STEP 4.
9. S2 SW CLOSED. S1 SW NORMAL.
10. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
11. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

FIGURE 12. Single-phase test diagram for leakage current measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.6.1 Failures. In addition to the definition of failure as specified in MIL-STD-721 and any definition specified in the applicable reliability test specification or standard, the criteria specified in 4.6.1.1 through 4.6.1.1.7 for the determination of a failure shall apply. Whenever any of the performance characteristics are outside of the requirements of the specification at any specified environmental conditions, at least one failure has occurred.

4.6.1.1 Relevant failures. All failures are relevant unless otherwise determined by the procuring activity. Relevant failures include those failures specified in 4.6.1.1.1 through 4.6.1.1.7.

4.6.1.1.1 Design or workmanship failures. Failures due to design deficiencies or poor workmanship of either the equipment or component parts shall be classified relevant.

4.6.1.1.2 Component part failures. Failures due to defective component parts shall be classified as relevant failures. In the event that several component parts of the same type fail during the test, each one shall be considered a separate relevant failure, unless it can be shown that one failure caused one or more of the others.

4.6.1.1.3 Wearout parts. Certain parts of known limited life such as batteries, may have a life stipulated prior to the initiation of testing as approved by the procuring activity. Failures of these parts occurring prior to the end of the stipulated period are relevant. Failures of these parts occurring after the stipulated period are nonrelevant, but any dependent failures caused thereby are relevant.

4.6.1.1.4 Mulitiple failures. In the event simultaneous part failures occur, each failed part which would independently prevent satisfactory equipment performance shall be counted as a relevant failure except as follows: If the procuring activity agrees that the failure of one part was entirely responsibile for the failure of any other parts, then each such dependent part failure shall not be counted as a relevant failure. At least one equipment relevant failure shall be counted when a dependent failure is claimed.

4.6.1.1.5 Intermittent failures. The first occurrence of an intermittent failure on any one equipment shall be counted as a revelant failure, although subsequent occurrences of the same intermittency on that same unit shall be considered nonrelevant. However, no equipment shall be shipped for which an intermittent failure remains unresolved, without specific approval of the procuring activity. If an intermittent failure masks the occurrence of a second failure; then, both shall be counted as relevant failures except if determined to be a dependent part failure in accordance with MIL-STD-781.

4.6.1.1.6 Adjustments (relevant failures).

4.6.1.1.6.1 Accessible controls. Each adjustment of a control which is accessible to the operator during normal use is a relevant failure if the information necessary to restore equipment operation is not available from indicators which are an integral part of the equipment under test.

4.6.1.1.6.2 Inaccessible controls. Each adjustment of a control which is inaccessible to the operator during normal use is a relevant failure, unless a period of operating (not standby) time has elapsed, since the last adjustment of any inaccessible control on that specific equipment or unit which is equal to or greater than the period specified for the equipment operational stability. This operational stability period shall not be interpreted as the system warm-up or performance stabilization period, and shall be identified in the applicable test procedure. Additionally, this time shall be not less than the minimum mean-time-between-maintenance-actions (MTBMA).

4.6.1.1.7 Mechanical failures. Mechanical disparities due to such tests as shock and vibration which prevent the system from successfully completing any other Government required test shall constitute a relevant failure.

4.6.1.2 Nonrelevant failures. The following failures may be counted as nonrelevant subject to final determination by the procuring activity:

a. Failures directly attributable to improper installation in test chamber

b. Failures of test instrumentation or monitoring equipment

c. Failures resulting from test operator error in setting up, or in testing the equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

d. Dependent failures, unless caused by degradation of items of known limited life (At least one relevant failure shall be counted when a dependent failure is claimed)

e. The second (and any subsequent) occurrence of the previously recorded intermittent failure on the same unit

f. Failures occurring during burn-in, trouble-shooting, repair verification, or set-up time if resolved prior to accrual of reliability test time or if so determined by the procuring activity

g. Failures clearly attributable to an overstress condition in excess of the specification requirements

h. Equipment failures caused by line voltage transients exceeding the maximum specified

i. Failures of the following items are considered nonrelevant during reliability tests:

1. Cables used for test but not part of system configuration

2. Printers and other peripheral I/O devices

4.6.1.2.1 Adjustments (nonrelevant failures).

4.6.1.2.1.1 Accessible controls. Adjustments of controls which are accessible to the operator during normal use shall be counted as nonrelevant failures provided the adjustment is accomplished without reference to test equipment, meters, indicators, and so forth, which are not a functional part of the equipment under test.

4.6.1.2.1.2 Inaccessible controls. Each adjustment of a control which is inaccessible to the operator during normal use is a relevant failure, unless a period of operating (not standby) time has elasped, since the last adjustment of any inaccessible control on that specific equipment or unit, which is equal to, or greater than, the period specified for the equipment operational stability. This operational stability period shall not be interpreted as the system warm-up or performance stabilization period, and shall be identified in the applicable test procedure.

4.7 Maintainability demonstration. Conformance with the requirements for equipment maintainability M_{ct} and M_{maxct} shall be determined by a maintainability demonstration in accordance with the general procedures of MIL-STD-471. The demonstration, corresponding to organizational level corrective maintenance, shall consist of 30 corrective maintenance tasks, each requiring replacement of a failed subassembly or chassis-mounted part.

4.7.1 Selection of corrective maintenance tasks. For the equipment maintainability demonstration, 60 candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471 and assigned numbers of 1 to 60. The following information shall be made available for each candidate task:

a. Designation of specific failed subassembly and failed part within the subassembly or failed chassis-mounted part

b. Failure mode

c. Means of introducing failure (substitution of faulty part or simulation thereof)

The procuring activity or its representative shall use the 60 candidate tasks as a guide to select a sample of 30 tasks for the demonstration.

4.7.2 Preventive maintenance. All preventive maintenance tasks shall be performed to verify that the tasks may be accomplished easily with the instructions provided.

4.7.3 Technical documentation. Technical documentation for the demonstration shall be limited to the equipment technical manual. If the final manual is not available, a draft may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.7.4 Technician(s). The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration. The contractor shall provide sufficient instruction to qualify the Military technician(s) to perform the demonstration.

4.7.5 Accept or reject criteria. An accept decision for the maintainability demonstration requires that no more than 12 of the 30 corrective maintenance tasks exceed 20 minutes, and no more than one of the 30 corrective maintenance tasks exceed 40 minutes. The following procedures and data shall be utilized:

a. Maintainability. The time required to complete each corrective maintenance task shall be recorded as being within or in excess of the specified index (M_{ct} or M_{maxct}). The actual time shall also be recorded.

b. Self-test and failure isolation. After the introduction of each simulated failure or malfunction, the self-test circuits shall be exercized and during the demonstration, the isolation procedures described in the technical manual, together with the knowledge of the technician shall be used to isolate failures. Failure to detect a failure or correctly isolate the failure to a single subassembly shall be noted. If the failure cannot be isolated to a single subassembly or part, the ability to isolate to a group of three or fewer subassemblies or parts shall be recorded.

4.7.5.1 Rejection. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies, and

b. Upon approval of an approach to the redesign or corrections, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

4.8 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for shipboard use as a part of the AIMS MARK XII IFF System.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of first article samples to be submitted (see 4.3 and 6.3)

c. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)

d. Levels of preservation, packaging, packing and marking required (see 5.1)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of two units unless otherwise specified (see 4.3 and 6.2). The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC
(Project 5840-N207)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-28856A(EC)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Defense Standardization Program Branch
Department Of The Navy
Washington, D.C. 20360
Attn: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.